🧪 PHYSICAL & CHEMICAL CHANGES, SOLUTION — FULL MIND MAP

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📑 Topics in this chapter:

1. Physical vs Chemical Changes & Reaction Rate

- Physical change mein substance ka form/state badalta hai, chemical composition nahi. Water mein sugar dissolve karna reversible physical change hai; evaporation se sugar recover ki ja sakti hai. [Q1]
- Water ka liquid se vapour banna physical change hai, kyunki H_2O ki chemical identity same rehti hai; sirf state change hoti hai. [Q2]
- Chemical change mein molecular composition badalti hai aur new substance banta hai. Vegetables ka cook/soften hona irreversible chemical change ka example hai. [Q3]
- NaCl crystallization aur ice melting physical changes hain; milk souring chemical change hai, kyunki lactose lactic acid mein convert hota hai. [Q4]
- Fruit ripening, cement setting aur coal burning tino chemical changes hain aur broadly exothermic nature show karte hain; source mein "no odd one" diya hai. [Q5]
- Higher temperature par molecules ki kinetic energy aur collision frequency badhti hai, isliye chemical reaction generally faster hoti hai. [Q6]
- Reaction/decomposition context mein energy change reaction type par depend karta hai; source Q7 "heat" key karta hai, lekin wording ko carefully read karna chahiye. [Q7]

> **⚠️ EXAM TRAP — Q7 — "Decomposition of Water" Wording**
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> ▸ Agar question actual electrolysis/decomposition of water ki baat kare, to electrical energy input hoti hai. Source explanation hydrolysis se mix ho rahi hai aur "heat produced" key scientifically context-dependent/ambiguous hai. [Q7]

2. Pasteurization & Phosphatase Test

- Pasteurization mein milk ko controlled temperature tak heat karke specified time ke baad rapidly cool kiya jata hai, jisse pathogenic microbes drastically reduce hote hain aur shelf-life improve hoti hai. [Q8]
- Milk pasteurization ko exam options mein "sterilization" se relate kiya ja sakta hai, lekin scientifically ye complete sterilization nahi, controlled heat-treatment / partial microbial reduction hai. [Q9]
- Desalination–sea water, reverse osmosis–potable water aur denaturation–proteins sahi matches hain; pasteurization–tea standard match nahi hai. [Q10]
- Phosphatase test ka standard dairy use ye check karna hai ki milk pasteurization adequate hua ya nahi. Source key wider "all of the above" deti hai. [Q11]

⚠ EXAM TRAP — Pasteurization ≠ Complete Sterilization

▸ Pasteurization pathogens ko reduce/inactivate karti hai; sari microorganisms/spores ko necessarily destroy nahi karti. Isliye “sterilization of milk” exam-option shorthand hai, exact scientific definition nahi. [Q9]

▸ Phosphatase test ka established dairy use pasteurization adequacy check karna hai; source ka Tea/Water tak “all above” answer exam-key specific maana jaye. [Q11]

3. Boiling / Freezing Point & Solution Concentration

- Non-volatile impurity add karne par liquid ka vapour pressure kam hota hai aur boiling point increase hota hai — ise boiling-point elevation kehte hain. [Q12]
- Ice + salt freezing mixture 0°C se neeche temperature de sakta hai, kyunki salt freezing point ko lower karta hai; “salt freezing point badhata hai” statement false hai. [Q13]
- Solution ki concentration = given amount/volume/mass of solution ya solvent mein present solute ki quantity ka measure. [Q14]
- Ideal solution ke liye $\Delta H_{mix} = 0$ hota hai aur solution Raoult’s law follow karta hai; A–B interactions roughly A–A/B–B ke comparable hote hain. [Q15]
- Salt jaise non-volatile solute se water ka boiling point badhta aur freezing point ghatta hai; methyl alcohol jaise lower-boiling volatile component se mixture ka boiling point reduce ho sakta hai. [Q16]

4. Ideal Solutions & Electrolyte Conductivity

- Strong electrolyte ki specific conductivity (κ) dilution par generally decrease hoti hai, kyunki ions per unit volume kam ho jate hain; molar conductivity ka trend alag hota hai. [Q17]

⚠ EXAM TRAP — Conductivity vs Molar Conductivity

▸ Dilution par specific conductivity κ decrease hoti hai, lekin molar conductivity Λ_m generally increase hoti hai. Exam mein “conductivity” kis quantity ko refer kar rahi hai, wording dekhein. [Q17]

5. Colloids, Emulsions & Homogeneous Solutions

- Milk oil-in-water type emulsion hai — fat droplets aqueous phase mein dispersed rehte hain; emulsion colloid ka ek type hai. [Q18, Q19]
- Milk, blood aur ice-cream colloidal systems ke examples hain; source ke according honey colloid nahi hai. [Q20]
- Fog aerosol type colloid hai: tiny liquid water droplets gas/air mein dispersed rehti hain. [Q21]
- Homogeneous solution ki composition throughout uniform hoti hai; salt-water homogeneous hai. “Different composition throughout” heterogeneous system describe karta hai. [Q22]
- Source Q23 unsaturated sugar solution mein volume ko unchanged key karta hai; practical solution volume exact additive/unchanged maana nahi ja sakta. [Q23]

⚠ EXAM TRAP — Sugar Solution Volume — Source Simplification

▸ Solute dissolve karne par solution volume generally exactly initial solvent volume ke equal nahi rehta; molecular interactions ki wajah se contraction/expansion ho sakta hai. Source Q23 ka “volume unchanged” idealized/exam-key treatment hai. [Q23]

6. Osmosis & Reverse Osmosis

- Plant roots soil se water mainly osmosis ke through absorb karti hain — semipermeable membrane ke across solvent lower solute concentration se higher solute concentration side move karta hai. [Q24]
- Cucumber par salt lagane se outside solution hypertonic ho jata hai aur cell water bahar nikalta hai; ye osmosis/exosmosis ka example hai. [Q25]

- Normal osmosis mein solvent dilute solution → concentrated solution jata hai. Reverse osmosis mein concentrated side par external pressure apply karke flow reverse kiya jata hai. [Q26]
- Pure solvent aur solution ke beech semipermeable membrane ho to solvent naturally solution side jata hai; ise osmosis kahte hain, reverse osmosis nahi. [Q27]

> ⚠ **EXAM TRAP — Osmosis Direction**
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> ▸ Osmosis mein solvent dilute/low-solute side se concentrated/high-solute side jata hai. Reverse osmosis mein pressure concentrated side par apply hota hai — dono direction statements frequently reverse pooche jate hain. [Q26, Q27]

7. Separation & Purification Techniques

- Distillation mainly liquid mixtures/liquid purification ke liye useful hai; solid purification mein sublimation aur crystallization common techniques hain. [Q28]
- Chromatography mixture ke components ko stationary aur mobile phases ke beech differential distribution ke basis par separate karti hai. [Q29]
- Camphor ka solid se directly vapour mein jana sublimation hai — beech mein liquid phase nahi aati. [Q30]
- Camphor, naphthalene aur dry ice jaise substances solid → gas direct transition dikhate hain; phenomenon sublimation kehlata hai. [Q31, Q32]
- Sand + naphthalene mixture ko sublimation se separate kiya ja sakta hai: naphthalene sublime karta hai, sand nahi. [Q33]

8. Oxidation Number

- Oxidation-number matching: Mn in MnO_2 = +4; S in H_2SO_4 = +6; Ca in CaO = +2; Al in $NaAlH_4$ = +3. Isliye A-3, B-4, C-1, D-2 mapping banti hai. [Q34]

9. Catalysts & Reaction Energy

- Catalyst reaction rate increase karta hai aur promoter catalyst ki activity enhance karta hai. Reaction ko slow karne wale substances inhibitors/negative catalysts kehlate hain; given options mein none correct tha. [Q35]
- Exothermic process heat release karta hai: concentrated H_2SO_4 ka dilution aur quicklime (CaO) + water reaction exothermic hain. Water evaporation aur camphor sublimation endothermic hain. [Q36]

10. Displacement Reaction & Corrosion

- Displacement reaction mein more reactive metal less reactive metal ko uske salt solution se displace karta hai. Cu, $AgNO_3$ se Ag ko displace karke copper nitrate + silver banata hai. [Q37]
- Corrosion metals ka gradual chemical deterioration hai: iron par brown rust, copper par green patina aur silver par black tarnish tino corrosion ke examples hain. [Q38]